FOR ITE153-AI AND EXPERT SYSTEM

2026

Research Project Information System with Rule-Based Decision Support

PRESENTED BY

BAGARES, XARZHA MAURINE
LUNGAY, VANISA G.
PAMISA, EJ KRISHIA ANN F.

Problem Overview

Many educational institutions still rely on manual or partially digital methods in managing research proposal submissions, which often result in delays, inconsistent checking of requirements, poor record organization, and difficulty in monitoring proposal status. The manual verification of documents also increases staff workload and the possibility of human errors such as overlooked or incomplete requirements. In addition, the lack of a centralized system makes communication and coordination among proponents, reviewers, and administrators less efficient. To address these challenges, this study proposes a Research Project Information System with Rule-Based Decision Support that automates requirement checking using predefined IF-THEN rules, helping improve the efficiency, accuracy, consistency, and reliability of research proposal management.

Research Project
Information System

AI / Expert System Concept

- Uses a Rule-Based Expert System under Artificial Intelligence and Knowledge-Based Systems
- Applies IF-THEN production rules to evaluate proposal submissions
- Automatically checks if all required documents and proposal sections are uploaded
- Generates “Ready for Review” status when all requirements are complete
- Generates “Incomplete” status when one or more requirements are missing
- Identifies and displays missing submission requirements automatically
- Reduces manual checking and staff workload
- Improves evaluation speed, accuracy, and consistency
- Minimizes human errors in validating proposal requirements

SYSTEM DESIGN

The proposed system follows a client-server architecture composed of frontend, backend, and rule evaluation components to support efficient management of research proposal submissions. The frontend is developed using HTML, CSS, and JavaScript to provide user interfaces for submitting proposal information, uploading documents, and monitoring submission status, while the backend uses Node.js and Express.js to handle request processing, file validation, and rule-based evaluation.

- **Dashboard Module –** displays overview information and proposal statistics
- **Proposal Submission Module –** allows users to input proposal details and upload required documents
- **Rule-Based Evaluation Module –** evaluates submissions using predefined IF-THEN rules
- **Submission Monitoring Module –** displays proposal records and submission status

Algorithm / Reasoning Method

The system uses a Rule-Based Algorithm based on Forward Chaining Reasoning to evaluate research proposal submissions using predefined IF–THEN rules that determine whether requirements are complete or incomplete. The evaluation process starts when a user submits proposal data and uploads required documents, which are then validated and processed by the system.

- It applies forward chaining by matching submitted facts against rule conditions
- When a condition is satisfied, the corresponding action is automatically triggered
- The system evaluates all rules and stores the results for each requirement
- It computes the overall completion status of the proposal

The final decision follows this logic: IF all rules are satisfied → "Ready for Review", otherwise IF one or more requirements are missing → "Incomplete." This approach ensures automated, consistent, and efficient evaluation while reducing manual checking and human errors.

DEMO

RESULTS

The developed Research Project Information System successfully automated the evaluation of research proposal submissions using rule-based decision support. The system was able to:

- manage proposal submissions digitally
- identify missing proposal requirements
- automatically generate submission status
- calculate completion percentage
- provide diagnostic feedback to users

The implementation of the rule-based evaluation mechanism improved consistency and reduced the time required for manual checking of proposal requirements. The system also improved organization of proposal records and provided a more structured workflow for proposal management.

CONCLUSION

The study successfully developed a Research Project Information System with Rule-Based Decision Support that improved the management, evaluation, and monitoring of research proposal submissions through automated and organized digital processes. The system provided a centralized platform for submitting proposals, uploading required documents, and tracking submission status efficiently.

- It integrated a Rule-Based Expert System using predefined IF–THEN rules
- Automatically determined whether proposals are “Ready for Review” or “Incomplete”
- Ensured consistent and structured evaluation of submissions
- Reduced manual workload and minimized human errors in checking documents
- Provided feedback on missing requirements for easier correction and resubmission

Overall, the system demonstrated that Rule-Based Decision Support is an effective AI approach for automating research proposal evaluation, improving efficiency, accuracy, consistency, and reliability in managing institutional research processes.

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thank you

Bagares, Xarzha Maurine

Lungay, Vanisa G.

Pamisa, Ej Krishia Ann F.